Episodic Precipitation

T. Lapp,[1,2] M. Rohloff,[1,2] M. Wilkinson,[3] and J. Vollmer[1,2]

[1]*Max Planck Institute for Dynamics and Self-Organization (MPI DS), 37077 Göttingen, Germany*
[2]*Faculty of Physics, Georg-August Univ. Göttingen, 37077 Göttingen, Germany*
[3]*Department of Mathematics and Statistics, The Open University, Walton Hall, Milton Keynes, MK7 6AA, England*

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Episodic outbreaks with potentially catastrophic consequences are characteristic of hot-[1,2] and cold-water[3] geysers, as well as lake[4,5] and volcano[6,7] eruptions. These processes have in common that a slow continuous mass or heat flux into a fluid mixture leads to bubble nucleation and growth, and that episodically the material accumulated in the bubbles is released in explosive precipitation events. Analogous modulations of precipitation rates also arise in laboratory experiments where phase separation in binary fluids can be monitored during pressure release[8] or a temperature ramp.[9,10] In all settings the description of the frequency of the episodic precipitation is a problem of vivid interest.[2,3,6,11] Here we show that the time scale, Δt, separating subsequent outbreaks is selected by a bottleneck arising at the cross-over from diffusive to explosive bubble growth. We obtain a parameter-free prediction of Δt by pinpointing the thermodynamic parameters that govern the time scale required to pass the bottleneck. This prediction is in a very good quantitative agreement with experimental data on two dedicated experiments where the temperature protocol is varied over a vast range of parameters. Based on these findings we revisit episodic precipitation in geophysical settings.

In our experiment, Fig. 1, two partially miscible liquids form two layers with a phase which is richer in the less dense fluid floating over a layer of the high-density phase. A 5 mL light-scattering cell containing this mixture is placed in a water bath where its temperature is varied smoothly away from the phase coalescence point, T_c, and the time-dependence of the temperature is engineered so that the rate of change, ξ, of the droplet volume fraction remains constant in each run of the experiment. Both layers show an alternating variation in turbidity, Fig. 1.a–f. Representing this evolution in a space-time plot, Fig. 1.g, illustrates an approximately periodic variation of turbidity. The alternation in the turbidity, that is characteristic of episodic precipitation, is also reflected in the particle size distribution[10] and in calorimetric data.[9,12,13] The effect is robust. It is observed in a vast range of binary mixtures,[9,10,12,13] including olive oil and methylated spirit,[11] and it arises in the upper as well as in the lower phase of the mixtures. In this respect it is appealing to establish a universal description of the oscillations, and to explore what insight this might provide in episodic precipitation arising in the geophysical settings.

As a first step to discuss Δt we consider the reasons why the turbidity — and hence the precipitation rate — in our experiment is not steady: the turbidity of a transparent solution increases when a considerable number of droplets have grown to a size comparable to (and eventually larger than) the wavelength of light. This manifests as a change of colour in the lower part of the cell when the system progresses from the snapshots shown in Fig. 1.b to c. Conversely, the solution becomes clearer again when vast amounts of small droplets are collected during the sedimentation of the largest droplets (transition from Fig. 1.d to e). Repetition of the cycle of nucleation, growth of droplets, and resetting the system by sedimentation gives rise to episodic precipitation, as shown in the space-time diagram, Fig. 1.g. In the following the salient features of this dynamics are modelled. We base our analysis on the growth and sedimentation of the largest droplets in the system.

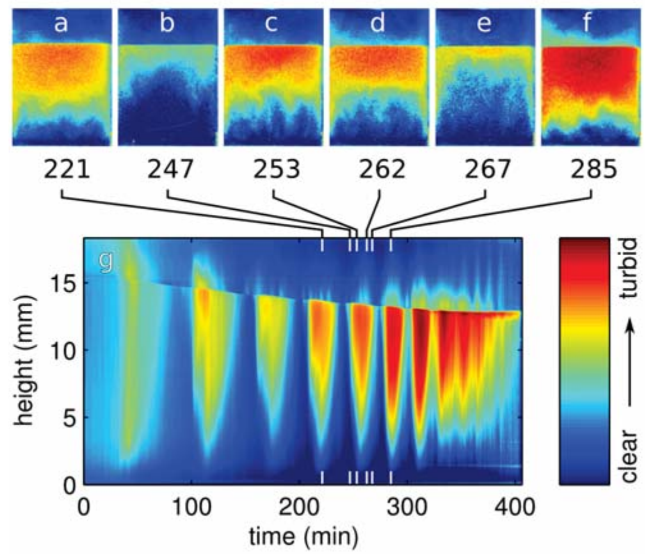

FIG. 1. **Evolution of turbidity**. Panels a–f show false-colour plots of the turbidity distribution in typical snap shots of the phase separation of an isobutoxyethanol/water mixture subjected to a ramp rate of $\xi = 2.5 \times 10^{-5}\,\mathrm{s}^{-1}$. Averaging in horizontal direction and arranging the resulting vertical turbidity profiles next to each other produces a space-time plot of the time evolution of the turbidity, panel g. The length scale is provided on its ordinate axis, and the scales in the pictures a–f an be inferred by noticing that panel g shows the full height of the samples, which have a square cross section. In the supplementary information we provide full details on the experimental setup and method, and we supply movies of the full evolution, as provided by our camera and in the false colours presented here.

Supersaturation causes nucleation of microscopic droplets in the solution. Subsequently, the supersaturation in the bulk relaxes by diffusion of the minority component onto the

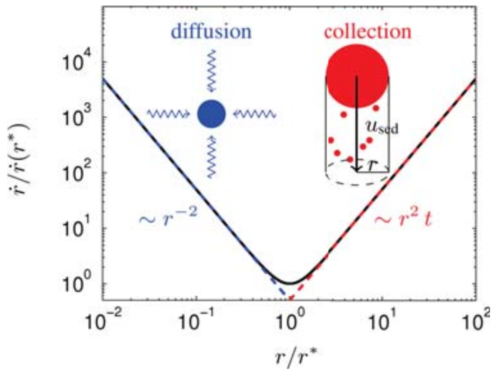

FIG. 2. **Bottleneck in droplet growth**. As a function of droplet size r the growth speed $\dot{r}$ of droplets shows a sharp minimum at a size r^*. Smaller droplets grow by diffusion — for larger droplets sedimentation speeds up growth by promoting the collection of small droplets (see insets).

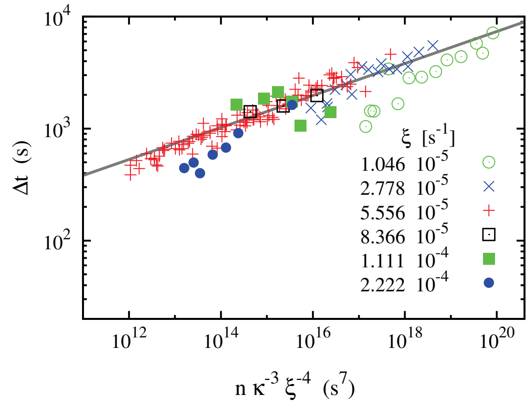

FIG. 3. **Verification of Eq. (5).** The period of episodic precipitation in isobutoxyethanol/water mixtures of various compositions collapse on a curve when Δt is plotted as a function of $n\kappa^{-3}\xi^{-4}$. Here, the respective heating rates in units of s^{-1} are specified in the legend of the plot and the measurement of the densities have been reported in Ref. 10. When collecting the numerical prefactors and accounting for the temperature dependence of κ, the collection efficiency, ϵ, remains as the only free parameter of the prediction. It induces a vertical displacement of the line, and was set to the value $\epsilon = 0.3$ for the line shown by the solid curve.

droplets, causing them to grow. While many different processes contribute to the growth, we demonstrate now that it suffices to consider only two processes in order to achieve a quantitative description of Δt. The processes are illustrated in Fig. 2:

1. The temperature ramp maintains a considerable supersaturation in the mixtures. In the experiments it is adjusted in such a way that the volume fraction of droplets grows linearly in time with a rate ξ. Droplets of a characteristic radius r and number density n occupy a volume fraction $n\, 4\pi r^3/3$. In this regime the number of droplets changes slowly as compared to their growth rate, $|\dot{n}\, r^3| \ll |n\, r^2\, \dot{r}|$, and the rate of increase of the droplet volume fraction, ξ, leads to a diffusive growth rate

$$\dot{r} = \frac{\xi}{4\pi n r^2}\,. \tag{1}$$

2. When the droplets become sufficiently large, they will drift under the influence of buoyancy forces. According to Stokes' formula the velocity of a slowly settling sphere is $u = \kappa r^2$, where κ is a material constant involving the gravitational acceleration, the viscosity, and the densities of the respective phases. When the Stokes velocity of the the largest droplets in the system becomes noticeable they sweep up the smaller droplets in their path. Hence, the volume of a large droplet grows at a rate $\epsilon\pi r^2 u\xi t$, where ϵ is the collection efficiency for large droplets coalescing with smaller ones and ξt is the volume fraction of the smaller droplets. This leads to a collisional growth rate

$$\dot{r} = \frac{\epsilon\kappa\xi t}{4} r^2\,. \tag{2}$$

The diffusive growth mechanism, Eq. (1), works very well for small droplets due to the factor r^{-2}, and it becomes less and less efficient when r grows. On the other hand, the collisional sweeping mechanism, Eq. (2), does not contribute to the growth of small droplets while it leads to runaway growth of the largest droplets. Hence, we assert that the sum of the diffusive growth, Eq. (1) and the sweeping contribution, Eq. (2),

$$\dot{r} = \frac{\xi}{4\pi n}\frac{1}{r^2} + \frac{\epsilon\kappa\xi t}{4} r^2\,, \tag{3}$$

provides the growth rate of the largest droplets in the system. This growth law shows a bottleneck of growth at the time t^* where $\dot{r}$ takes its minimum (see figure 2),

$$r^* = r(t^*) \simeq \left[n\epsilon\kappa\, t^*\right]^{-1/4}\,. \tag{4}$$

Integrating Eq. (3) for an initial condition $r(t=0) = 0$ yields that the droplet radius diverges when growth proceeds for a *finite* time proportional to t^*. (This is an immediate consequence of non-dimensionalisation of Eq. (3), and observing that t^* is the only time scale in the problem — a formal derivation is provided in the supplementary information.) Consequently, the time scale, t^*, provides a faithful description of the parameter dependence of the time scale Δt between subsequent precipitation events.

Combining Eq. (4) with $(r^*)^3 \sim \xi t^*/n$, which results from integrating Eq. (1) from $r(t=0) = 0$ to $r(t^*) = r^*$, we find

$$\Delta t \sim t^* \sim \left(\frac{n}{\kappa^3\xi^4}\right)^{1/7} \tag{5}$$

with a pre-factor of order one. Besides the material constant κ and the control parameter ξ this prediction only depends upon the value of the droplet density n. Based on the droplet size

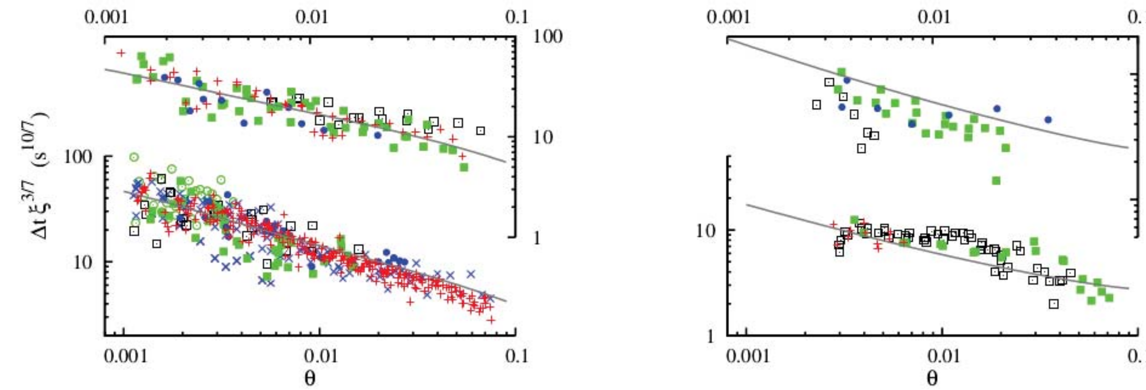

FIG. 4. **Temperature dependence of $\Delta t\, \xi^{3/7}$ for isobutoxyethanol/water and methanol/hexane mixtures**. Data points are shown for the upper (top) and the lower layer (bottom) of mixtures of isobutoxyethanol/water (left) and methanol/hexane (right), respectively. The respective solid lines are the theoretical prediction, Eq. (6)), with $\alpha = 0.71$ (left) and $\alpha = 0.9$ (right). The colours and symbols encode different heating rates ξ: green circle, $\xi < 6 \times 10^{-6}\,\mathrm{s}^{-1}$; blue cross, $10^{-6}\,\mathrm{s}^{-1} < \xi < 1.3 \times 10^{-5}\,\mathrm{s}^{-1}$; red plus, $1.3 \times 10^{-5}\,\mathrm{s}^{-1} < \xi < 3 \times 10^{-5}\,\mathrm{s}^{-1}$; black square, $3 \times 10^{-5}\,\mathrm{s}^{-1} < \xi < 6 \times 10^{-5}\,\mathrm{s}^{-1}$; green squares, $6 \times 10^{-5}\,\mathrm{s}^{-1} < \xi < 3 \times 10^{-4}\,\mathrm{s}^{-1}$; and blue circle, $3 \times 10^{-4}\,\mathrm{s}^{-1} < \xi$. In response to the increase of θ the period typically drops by an order of magnitude, ranging between a few minutes for the largest values of ξ till several hours for small values of ξ and θ. (All material constants and the raw data are provided in the supplementary information.)

distribution reported by Lapp *et al*[10] we demonstrate in Fig. 3 that the prediction provides a very good description of Δt.

Typically the droplet density, n, is not easily accessible. It is therefore desirable to provide an estimate for n in order to arrive at a widely applicable theory for the period, Δt. In the context of the growth of nano-particles it has been established[14,15] that when the particle volume fraction is growing in time, $\xi > 0$, the number density n is proportional to $\xi/(\sigma D)$ where D is the pertinent diffusion coefficient for the accretion of material on the droplets, and σ the Kelvin length, which takes values of the order of the interface thickness. Combining this proportionality with Eqs. (5) we obtain

$$\Delta t = \alpha \left(D\sigma\kappa^3\right)^{-1/7} \xi^{-3/7} \,. \tag{6}$$

The coefficients D, σ and κ are functions of material constants. They show a strong temperature dependence, which arises from the vanishing of the interfacial tension and the mass density contrast at the critical temperature, T_c, of the phase transition. (This, in turn, entails the vanishing of σ and κ which are proportional to the interfacial tension and the mass density contrast, respectively.) Most conveniently the temperature dependence of Δt is therefore specified in terms of the reduced temperature $\theta = |T - T_c|/T_c$. Together with Eq. (6) this argument suggests that $\Delta t\, \xi^{3/7}$ should be a function of θ. This expectation is corroborated by the data collapse in Fig. 4 where we have collected data of Δt for a vast range of heating rates ξ, and four different scenarios of phase separation in a binary mixture: (top left) the emergence and sedimentation of water-rich droplets in an isobutoxyethanol-rich phase; (bottom left) the emergence and rising of isobutoxyethanol-rich bubbles in a water-rich phase; (top right) the emergence and sedimentation of methanol-rich droplets in a methanol-rich phase; and (bottom right) the emergence and rising of hexane-rich bubbles in a hexane-rich phase.

The dimensionless prefactor α is the only free parameter in Eq. (6). This parameter takes values very close to unity that only depend on the selected mixture: $\alpha = 0.71$ for isobutoxyethanol/water (left panels of Fig. 4), and $\alpha = 0.9$ for methanol/hexane (right panels of Fig. 4). Consequently, the θ dependence of Δt is faithfully provided by the temperature dependence of the material constants in Eq. (6), as demonstrated by the solid lines in Fig. 4.

The faithful description of episodic precipitation displayed in Fig. 4 reflects the interplay of diffusive droplet growth and a runaway instability of droplet size that arises when the largest droplets start to be effected by gravity. The runaway crucially depends on the assumption of a polydisperse droplet distribution. Its treatment has been inspired by models for rain initiation in warm clouds.[16]

State of the art models of geyser and volcano eruptions rather involve two-fluid models that account for the reduction of density due to changes of the bubble volume fraction and its changes in response to pressure release. They do not consider the bubble size distribution.[1–3,6] The mechanism of episodic precipitation contributes an additional oscillatory instability in their dynamics that generates similar periods as the pressure based models (the data shown in Fig. 4 exhibit periods ranging from 10 minutes to half a day). This calls for follow up modeling and experimental work in order to discriminate both effects in these applications. In which situations would Eq. (6) fully account for the period of the episodic outbreaks? Can episodic precipitation interact non-trivially with the pressure feedback in order to give rise to the highly irregular time series of the outbreaks in geysers and volcanoes?

The rates and kinetics of bubbles (or light particles) rising, growing, merging, and dissolving in deep water explicitly depend on the size distribution. Hence, our present results on episodic precipitation suggest that the droplet size distribution must explicitly be accounted for in theories addressing the arrest of lake and ocean eruptions[5] due to bubble dissolution,[17] or the suppression of outbreaks due to the formation of a solid clathrate phase in the case of methane.[18]

Another intriguing area of applications emerges when acknowledging that the bottleneck of growth, that underlies episodic precipitation, is also a key factor in warm rain initiation[16,19,20] which is a major outstanding problem in current climate research.[21] In this setting the identified parameter dependence provide a guide line for improving parametrisation schemes of rain initiation, in particular in numerical simulations resolving the lateral extension of clouds.[22] Moreover, the prediction of the time scale for rain initiation should also apply to exotic types of rainfall in exoplanetary atmospheres[23,24] because it is not tailored to specific atmospheric conditions and compositions.

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